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**OJTPORTAL: A WEB-BASED APPLICATION OF ON-THE-JOB
TRAINING MONITORING AND INFORMATION SYSTEM
OF THE POLYTECHNIC UNIVERSITY OF THE
PHILIPPINES INSTITUTE OF TECHNOLOGY**

A Capstone Project

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

The OJT (On-The-Job Training) season in Polytechnic University of the Philippines–Institute of Technology (PUP ITECH) happens every summer term for 2nd year and every 2nd semester for 3rd year students. The school's OJT system is currently being handled through manual processes such as submitting hard copies of documents to the school and requesting wet signatures from coordinators and advisers.

However, with a high volume of students and documents being checked and submitted, administrative workload processes such as document processing, approval workflows, and timely communication with stakeholders often becomes overwhelming and inefficient as tons of paperwork are being collected. While there are efforts to mitigate such gaps in the process, the common problems, not limited to the previously mentioned aspects, continue to burden the stakeholders involved.

By developing a streamlined OJT portal system, this project aims to address these challenges and improve the overall experience. The objective of the OJT Portal is to automate processes for submissions, track students' progress by a status system, and provide real-time updates for every process during OJT. Through an organized and automated process, students can focus more on work preparation rather than on the lengthy process of document submission and approvals. Administrative workloads can also be reduced, thus helping the school administrators, OJT coordinators, professors, and HTEs to manage tasks and time better.



Theoretical Framework

Developing a new system to be adopted by an institution can improve certain issues it would aim to address. In this study, the proposed OJTPortal intends to alleviate concerns when it comes to the hassle in document submission, appointment scheduling, and other related processes for internship preparation. However, it is important to take note that while the proposed system will be constructed with positive impact in mind, the intended users may be reluctant to accept a new system as they are accustomed to the traditional method of submitting and processing hard-copy documents and other requirements.

The Technology Acceptance Model (TAM) by Davis (1989) serves as a foundation in determining and understanding the factors that may affect how the users are influenced in accepting the proposed system. The main determinants in the model are the Perceived Usefulness and Perceived Ease of Use, which both have significant impact on the attitude of the user when it comes to the usage intention of a new technology. The usefulness of a system perceived by the user refers to “the degree to which a person believes that using a particular system would enhance his or her job performance” (Davis, 1989, p. 320). The perceived ease of use, on the other hand, refers to “the degree to which a person believes that using a particular system would be free of effort” (Davis, 1989, p. 320). It is claimed that these two determinants have a causal effect on adopting or rejecting a new technology, and the higher the perceived usefulness and ease-of-use of the system, the more likely it is to be used by the user.

In the context of this study, the independent variables are the quality presented in the OJTPortal such as the overall functionality, accessibility, design, and



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performance, as these are expected to influence the perceived usefulness and ease of use of the system in relation to the TAM theory. The dependent variables include the outcomes such as the efficiency of different OJT processes, the user satisfaction, and the effectiveness and adoption in the actual use of the portal.

Furthermore, the study will also evaluate the system's quality in accordance with the ISO/IEC 25010 model to assess the six out of nine main characteristics of a quality software: functional suitability, compatibility, interaction capability, maintainability, flexibility, and safety. The researchers carefully selected these specific characteristics while keeping in mind the possibility of presenting only the prototype of the system upon sending out survey questionnaires.

By applying this software quality model, the researchers can measure the TAM determinants comprehensively. The characteristics of functional suitability, compatibility, maintainability, and safety can affect the perceived usability, while interaction capability can influence the perceived ease of use. On another note, ISO/IEC 25010 characteristic of flexibility can impact both TAM determinants. By combining TAM theory with the ISO/IEC 25010 model for software quality standard assessment, the study will be able to determine how the quality of the proposed OJTPortal influence user acceptance and overall effectiveness of the system to address the existing problems in the long-standing traditional process of OJT in PUP ITECH.

Conceptual Framework



Statement of the Problem

1. What are the perceived challenges faces by stakeholders in the traditional OJT process, and to what extend do these challenges affect the efficiency of OJT management?
 - a. Students
 - b. OJT Advisers
 - c. OJT Coordinator
 - d. HTE Representatives
2. How do stakeholders perceive the potential effectiveness of the OJTPortal prototype in addressing these identified challenges?
3. What is the level of acceptance of the respondents on the OJTPortal based on ISO/IEC 25010 in terms of:
 - a. Functional Suitability
 - b. Compatibility
 - c. Interaction Capability
 - d. Maintainability
 - e. Flexibility
 - f. Safety
4. How do stakeholders perceive the OJTPortal's potential contribution to the following Sustainable Development Goals (SDGs)?
 - a. SDG 4 – Quality Education
 - b. SDG 8 – Decent Work and Economic Growth
 - c. SDG 9 – Industry, Innovation, and Infrastructure
5. How do stakeholders perceive the feasibility and readiness for implementation of the OJTPortal in actual OJT processes?



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Scope and Limitations of the Study

This research focuses on developing a web-based OJT portal system to streamline and automate processes such as document management, DTR (Daily Time Record) tracking, real-time communication of stakeholders, appointment scheduling, MOA (Memorandum of Agreement) management, and notifications for status updates and messages. The primary users of the portal are students undergoing OJT, OJT coordinators, advisers, and HTE representatives. The project is limited to all undergraduate programs under PUP ITECH and is designed specifically for the needs of ITECH and may not fully meet the requirements of other institutions or programs.

The system will require an active internet connection to function as OJTPortal is not designed for offline functionality at this stage of the study. The project will also not include advanced analytics, AI (artificial intelligence) features, or any third-party software integration as the development is constrained by limited time and expertise. The portal will only be tested on limited sample users from one department(?) to validate its functionality and usability. Full deployment across all programs handled by ITECH is not part of this study.

Significance of the Study

This study wants to determine the effectiveness of the proposed OJTPortal in streamlining OJT management and related processes for PUP ITECH. The following sectors mentioned will benefit from this study:

- **Students**



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Students may benefit from the use of OJTPortal as they will be able to submit their OJT documents online, monitor the internship tasks and document approvals, and apply for appointments at school for medical clearance and other documents to be requested.

- **OJT Coordinators and Advisers**

OJT coordinators and advisers may benefit from the proposed OJTPortal as they can monitor the progress of students when it comes to the submission of requirements and the actual internship work being completed daily/weekly. The convenient messaging system may also benefit them as direct communication with concerned students or HTE will be easier.

- **HTEs**

The partner HTEs of ITECH may benefit from this proposed system as they can easily track and monitor the current student interns being trained in their company. The feedback and evaluations can easily be sent to the respective advisers to avoid delays in the process.

- **PUP**

This study only focuses on ITECH, but the PUP itself may benefit from this proposal as they can adapt the system to a wider reach of branches and campuses across the country, which then can help the university to have an advanced and comprehensive system for OJT.

- **Future Researchers**

Researchers may use this study as a guide or reference to develop a more advanced OJT portal with additional features such as AI or third-party software integrations. Researchers may also adopt other theoretical models to assess different aspects that can affect user satisfaction and system quality concerns.



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Definition of Terms

Adviser – This refers to a professor assigned to assist, guide, and monitor the progress of a specific students or section in the OJT period.

Compatibility – This is one of the quality characteristics under ISO 25010 which refers to the “degree to which a product, system, or component can exchange information with other products, systems, or components, and/or perform its required functions while sharing the same common environment and resources.”

Coordinator – This refers to the assigned school personnel that oversees all the OJT processes within the school and communicates with the respective HTEs and other school administrators.

Database – This refers to where all the data is stored and organized such as user information, documents, masterlist of students and HTEs, etc.

Flexibility – This is one of the quality characteristics under ISO 25010 which refers to the “degree to which a product can be adapted to changes in its requirements, contexts of use or system environment.”

Functional Suitability – This is one of the quality characteristics under ISO 25010 which refers to the “degree to which a product or system provides functions that meet stated and implied need when used under specified conditions.”

HTE - Host Training Establishment; This refers to a partner company of the school where students apply for their internship training

Interaction Capability – This is one of the quality characteristics under ISO 25010 which refers to the “degree to which a product or system can be



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interacted with by specified users to exchange information in the user interface to complete specific tasks in a variety of contexts of use.”

ISO/IEC 25010 -

Maintainability – This is one of the quality characteristics under ISO 25010 which refers to the “degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it, or adapt it to changes in environment, and in requirements.”

MOA - Memorandum of Agreement; This refers to a legal document that shows the written terms and conditions between the school and HTE when it comes to training student interns.

Safety – This is one of the quality characteristics under ISO 25010 which refers to the “degree to which a product under defined conditions to avoid a state in which human life, health, property, or the environment is endangered.”

SDG –



Chapter 2

REVIEW OF LITERATURES AND STUDIES

This chapter presents a review of literature and studies, both from local and international sources, related to the development and implementation of internship management systems. The thematic approach in discussing key topics related to the proposed OJTPortal establishes a foundation for innovation and contextualizes the need for a centralized, accessible, and efficient platform meant for higher education institutions in the Philippines.

Digital Transformation in Internship Management

Digital transformation has significantly reshaped higher education systems worldwide over the past few years. The demand has been influencing how institutions deliver programs, support student learning, and even prepare graduates for future employment and career paths. The improvement in the digitalization of such services is particularly evident in the management of OJT programs, which are crucial in bridging academic learning with real-world settings. Shifting to a digital format for internship management enhances efficiency, expands access to opportunities, and facilitates better monitoring and collaboration among stakeholders.

Globally, Rego et al. (2023) emphasized that digital transformation in education is a multidimensional shift that affects content delivery, student engagement, and employability. Raza and Hasib (2023) further supported this claim by demonstrating that integrating online tools and platforms into internship preparation and project-based learning significantly improved students' chances of securing meaningful internships. Tee et al. (2024) have also reported that the demand for digital and problem-solving skills among employers continues to grow,



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which reinforces the need for tech-driven internship models that cultivate such competencies early. These findings suggest that embracing digital strategies is no longer optional in the modern era but rather essential for innovative internship programs.

In the local context, a bibliometric analysis of technology integration in the Philippines has found that common barriers, such as time and distance, are addressed through digitalization. However, the reluctance to change and the adoption of modern technologies remain a challenge, along with minimizing the gaps in access, training, and support constraints (Cadiz et al., 2024). Climaco and Mangarin (2024) specifically pointed out that many Filipino students lack sufficient digital literacy due to limited access to technology and inadequate instructional support. The reality of the digital adoption status in the Philippines highlights the gap between global expectations for digital readiness and the current state of local internship delivery.

The study of Golhar et al. (2024) identified the role of internship portals in promoting inclusivity and accessibility in internship programs. Current technology overcomes geographical barriers and enhances visibility, and internship portals empower students from diverse backgrounds to access a broader range of internship opportunities. Additionally, centralized digital systems can streamline recruitment processes and help institutions manage internship activities more efficiently.

These studies emphasize that while digital transformation provides significant benefits in managing internships, its success in implementation is mainly dependent on user readiness, institutional support, and technological infrastructure. The proposed OJTPortal addresses this need by providing an accessible, web-based platform that streamlines progress tracking and communication among stakeholders. Through this, the system aims to bridge the existing digital divide and contribute to a more inclusive and effective internship process in the Philippines.



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Others:

Studies on digi transformation in educ and its impact on student employability

Case studies (foreign or local) on how internship platforms have changed recruitment and employer-intern engagement

Factors that affect the adoption of digi internship portals in universities

Digital portals vs traditional OJT methods in terms of efficiency and student experience

Features and Functionalities of OJT Portals

An appropriately designed OJT portal must go beyond the basic digital storage and serve as a functional system that streamlines the entire internship cycle, starting from the application and approval to monitoring and feedback. The effectiveness of an internship platform is often measured by its ability to simplify administrative tasks, improve user experience, and facilitate real-time tracking of intern progress.

Molodchik et al. (2020) analyzed 62 internship and job-matching platforms and then identified key functional categories such as application filtering, employer dashboards, and real-time status updates. It is also noted that specialized internship portals offer more comprehensive features than general job boards or university websites. Similarly, the World Bank (2023) emphasized the role of AI and data-driven platforms in providing personalized and bi-directional matching between students and employers, which then significantly improves internship outcomes favorable for both parties. These insights support the important of integrating automation and smart



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matching into OJT systems to better align student skill with industry needs and requirements.

In the local context, Del Rosario and Dela Cruz (2022) introduced an Internship Program Management System designed with lean management principles. This system incorporates modules for daily time records (DTR), activity logs, and grading for evaluations. The system was positively received by both students and faculty, with users reporting reduced workload and increased process efficiency. This aligns with the findings of Andrian and Massaty (2025) in their usability study in Indonesia, where it was revealed that internship systems with clear, accurate, and efficient interfaces significantly enhanced both student engagement and administrative coordination.

The functionality of OJT systems is further illustrated in Cabalbag and Ontiveros's (2024) Management Information System for On-the-Job Trainees (MIS-OJT) platform. According to the authors, CSU Aparri's paper-based system causes inefficiencies, data loss risks, and administrative burdens. Students must travel to submit reports, and advisers handle significant paperwork. To address these issues, MIS-OJT was developed, with the goals of implementing online submissions, automated grade computation, data analysis, and improved feedback system.

Similarly, switching from traditional paper-based processes to digital workflows has revolutionized education, particularly during challenges like the COVID-19 pandemic (Illao et al., 2023). Tools like Microsoft 365 and Power Platform enable the creation of simple yet effective systems, which then became the inspiration in implementing a Sharepoint-integrated portal for OJT processes. A survey of 173 trainees revealed its effectiveness, with 54.91% rating it "Best" and 33.53% as "Good." This shows how digital platforms like the OJT Portal enhance efficiency and improve the OJT experience for students and institutions.



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These developments highlight the benefits of structured workflows, document integration, and centralized access to internship-related tasks. By incorporating such features to the design of the proposed OJTPortal, the system not only supports institutional efficiency but also enhances the internship experience for student and partner HTEs.

Borra (2024)

Others:

Studies covering different internship management platforms worldwide

Role of AI-driven matching systems in internship placement

Essential components of an effective OJT portal

Impact of automation on student performance tracking and evaluation

Document Management and Verification Systems

Efficient document management is essential in OJT programs where various stakeholders handle time-sensitive and confidential records such as contracts, evaluation forms, and time logs. Traditional paper-based systems are prone to delays, data loss, and miscommunication. The shift to digital document management and verification systems aims to solve these inefficiencies through automation and security.

Blockchain and AI technologies are now being applied in academic credential verification globally. Noshi and Xu (2024) developed a blockchain-based system that uses QR codes for real-time validation of academic records, which significantly improves transaction speed and user trust. In similar settings, Cardenas-Quispe and



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Pacheco (2025) proposed a hybrid blockchain model that securely stores and verifies student information, therefore reducing the risks of tampering. Silaghi and Popescu (2023) reinforced these benefits by stating that blockchain's immutable ledger system helps to prevent fraud and ensures authenticity in academic documentation. These technologies enhance transparency, trust, verification speed, which are all essential for high-volume and document-intensive systems like OJT portals.

Locally, the implementation of Electronic Document Management Systems (EDMS) is perceived to be successful within university settings (Aliazas et al., 2024). Some of the benefits highlighted in the study are the improvements in efficiency, accessibility, and collaboration, given that potential challenges such as resistance to change and inadequate training and support are addressed properly and timely. Effective strategies for system implementation should be prepared thoroughly as those are crucial in ensuring positive user experience and providing high chances in continued and expanded usage of EDMS. Based on the findings in the same study, integrating such a system in key university processes significantly helps in overall effectiveness of implementation as streamlining administrative tasks are often desired by students and professors.

Another study by Abaci & Medeni (2022) pointed out in their paper that EDMS has a great impact on productivity, information communication, and quality of service. It allows the users to access information efficiently given that there are numbers of documents to be processed. This emphasizes how important time management means in their field and how it is saved with the use of EDMS. But results revealed that technological infrastructure and technical qualities are insufficient within the scope of the EDMS application. This suggests that it is important to improve or enhance the application in order to ensure the efficiency that the system offers and for it to be able to conduct the necessary R&D activities. Additionally, it was mentioned in the paper that some of the mistakes made while using EDMS are



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caused by human error. The solution to resolve this issue is to create standard templates at the system inputs, which led to the removal of personal preferences of employees do when submitting or handling documents.

Shende et al. (2024) supported these findings by emphasizing the use of machinelearning and image processing in document verification systems. In their review of such systems, the workflow automations necessary for the digitization of traditional paper-intensive processes will be key to driving increased effectiveness across a variety of industries, from education to the training environment. The use of technology such as image processing, machine learning, and electronic signature verification can help reduce human mistakes and repetitive work and achieve paperwork accuracy. The authors emphasize this will require creation of the integrated electronic territorial environment at the interface that will accommodate and safely authenticate documents for data integrity in addition to the capacity for a system that is experiencing large volumes of traffic in terms of users and transactions.

Much like the proposed OJT portal, these addresses the issues on document flow and interaction among students, professors, and Host Training Establishments (HTEs). The centralized platform makes this proposed OJTPortal compliant with the major findings outlined by Shende et al. (2024), particularly in the automation of the process, thus reducing time in action and enabling real-time tracking and feedback. Such integration of technology does not just help facilitate smooth workflow, but also instills visibility and accountability, which are critical elements upon which training and education systems are built.

Luyombye & Ndagire (2020)

Others:



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Studies on blockchain tech in student record management

Use of AI in docu validation and fraud prevention

How does EDMS impact admin workload and docu retrieval speed

Security concerns in digital docu management for internships

Enhancing Communication and Collaboration in OJT

Effective communication and collaboration between students, OJT advisers, OJT coordinators, and HTEs are central to the success of any OJT program. Delays, miscommunication, or lack of feedback often lead to frustration and reduced productivity. Digital platforms provide solutions by enabling real-time interaction, centralized messaging, and task coordination, which then improves the overall OJT experience.

As reported by several recent studies, communication breakdowns have been identified as one of the major weaknesses of traditional internship systems, particularly the lack of consistent engagement between students and employers. The study of Mydyti & Kadriu (2020) outlines the problems with managing the internships, such as ineffective communication and not able to keep track of which student is where and how far along the internship experience is. It suggests establishing an Internship Management System (IMS) to centralize communication and promote continuous engagement between students, businesses, and faculty. Similarly, Sophia (2025) explored the use of AI-powered chatbots in educational portals and found that this feature enhances student engagement by responding to inquiries instantly, guiding users through tasks, and offering timely assistance.



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Mobile accessibility is another factor that influences collaboration in the OJT setting. According to Anuyahong and Pucharoen (2023), mobile-friendly learning platforms significantly increase student engagement and academic performance by allowing users to interact with systems anytime and anywhere. Real-time feedback is also crucial as supported by Salimi and Mohammadzad (2024) through stating that quick-response mechanisms enables educators and supervisors to adapt the instructions and resolve issues as they arise, which therefore leads to improved learning outcomes and task completion.

As previously reviewed under the features and functionalities of an effective OJT portal, integrating Microsoft 365 and Power Platform tools enhances coordination within Ilao et al. (2023) OJT portal. By enabling document sharing, chat, and approval workflows via Teams and SharePoint, communication delays were minimized and collaboration among stakeholders improved.

These findings support the design of the proposed OJTPortal, which will include messaging functions, automated notifications, and feedback modules. The system will be improved later on for mobile use to ensure that students and supervisors can communicate and check internship progress regardless of location. Additionally, a notification system will provide reminders for upcoming tasks, evaluations, or submission to help users to stay engaged and informed throughout the duration of internship.

Borra (2024)

Others:

Studies on the role of chatbots and AI assistants in OJT portals

Impact of mobile-friendly internship portals on accessibility



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How does real-time feedback improve student learning during internships

Challenges in implementing cloud-based collab tools in OJT programs

Challenges and Barriers in Implementing OJT Portals

Digital platforms offer numerous advantages for internship management, but several challenges hinder the success of implementation. These barriers often include limited infrastructure, digital literacy gaps, cybersecurity threats, and resistance to technology adoption.

According to UNESCO GEM Report (2023), digital inequality remains a major barrier in education, especially in low-resource settings where students lack devices, connectivity, or digital skills. Even when platforms are available, learners from marginalized communities may be excluded from fully participating in online systems. This is similar to the report released by U.S. Department of Education (2022) where it stated that cybersecurity remains a critical issue, with frequent data breaches and ransomware attacks are being received in school systems. These threats can compromise sensitive student and institutional data, which highlights the need for strong security protocols in educational systems.

In the local context, Cadiz et al. (2024) noted that while digitalization of systems is now a trend, challenges such as insufficient training, lack of access, and administrative resistance still exist. Climaco and Mangarin (2024) also revealed that many Filipinos struggle with basic digital literacy, often due to the socioeconomic disparities and underdeveloped ICT infrastructures in schools. Clearly, it can be



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noted how both technological and human factors can delay the adoption and effectiveness of systems like OJT portals.

Rodriguez (2024) added that implementing advanced technologies such as cloud computing and data analytics is limited by poor infrastructure and lack of skilled personnel in the Philippines. Such systemic issues must be addressed for any digital internship platform to be widely adopted and sustained.

Understanding these challenges, the proposed OJTPortal is designed with accessibility and security in mind. The goal is to include features the offer responsive interface for users with limited hardware, simplified navigation for digital beginners, and secure log-in and data encryption to protect sensitive information. Training module and onboarding materials will also be provided to assist students and faculty in adapting to the system, with the idea of reducing resistance and ensuring smoother implementation.

Abaci & Medeni (2022)

(Blockchain, AI, cloud adoption)

Others:

More studies on digital literacy and training apps among faculty and students

Impact of socioeconomic disparities on digital platform adoption

Factors contributing to student and faculty resistance to OJT portals

How can univs ensure equal access to digital internship platforms (since mostly are scholars with limited resources)

Frameworks and Models for Evaluating OJT Portals



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The successful implementation of an OJT portal is not solely determined by its technical features but also by how well it is accepted, used, and evaluated by its intended users. Evaluation models and performance indicators provide frameworks to assess usability, effectiveness, and impact to ensure that the system meets institutional goals and user needs.

Globally, various models have been used to evaluate education platforms. Krumova et al. (2023) emphasized the importance of tracking key performance indicators (KPIs) such as log-in rates, task completion, time-on-task, and user feedback to assess the effectiveness of learning management systems. These metrics help administrators monitor platform engagement and identify areas for improvement. Andrian and Massaty (2025) applied the User Experience Questionnaire (UEQ) and found that performance, clarity, and engagement were KPIs in internship systems. It is further noted the systems must be both functionally effective and user-friendly to ensure long-term adoption.

Beyond usability, technology acceptance models are commonly used in educational systems. The Technology Acceptance Model (TAM) evaluates user perception based on usefulness and ease of use. On the other hand, the Unified Theory of Acceptance and Use of Technology (UTAUT) supports the former by incorporating social influence and facilitating conditions. These models help predict user behavior and system adoption. Studies show that digital confidence, perceived effectiveness, and institutional support strongly influence system success.

In the previously mentioned MIS-OJT platform by Cabalbag and Ontiveros (2024), TAM and ISO standards are used to evaluate the system, which reflected positive feedback from student and faculty including evaluations mentioning the improved documentation, real-time tracking, and user convenience. Aliazas et al. (2024) also used TAM to understand users' perception on how they accept and adapt to new technologies. Faculty, staff, administrators, and even students can assess the



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effectiveness of a certain system in streamlining key university processes. Findings show that faculty and students are more likely to accept and continue using the platform when it is intuitive and integrates well with existing processes. This model will be used when implementing the proposed OJTPortal to ensure its positive impact and contribution in allowing the university to pursue a data-driven internship and education system.

Li (2021)

Others:

Additonal studies on usability testing methods for educ platforms

Use of student and employer feedback surveys in system evaluation

KPI (key performance indicators) to be used to measure OJT portal success

Use of predictive analytics to improve OJT placement and student outcomes

Relevance to Sustainable Development Goals

The development of digital platforms for internship management aligns with several global and national objectives, particularly the United Nations' Sustainable Development Goals (SDGs). Specifically, the proposed OJTPortal supports the following SDGs: SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation, and Infrastructure). All of which emphasize inclusive education, youth employment, and digital innovation, and these goals are addressed by systems that connect academic institutions and various industries through technology.



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For SDG 4, the proposed portal promotes inclusive and accessible education providing students with a digital platform to connect academic learning with real-world experience. A report by UNESCO (2024) highlights digital tools as essential in ensuring education continuity and skill development. Structured internship programs also support SDG 4.4 by equipping youth with practical skills aligned to industry demands (Aggarwal & Mishra, 2024).

In relation to SDG 8, the proposed OJTPortal aims to facilitate a smoother transition from education to employment. Digital platforms locally accessible in the Philippines like LUNA Academy and TESDA's partnership with GCash have shown that integrating training with job placement helps boost youth employability and economic participation (BusinessMirror, 2021; iTechSolutionPH, 2025). Similarly, the OJTPortal also helps connect students with host companies, which then supports full and productive youth employment under SDG 8.5.

Aligned with SDG 9, the portal contributes to innovation in education systems by digitizing manual OJT processes and strengthening infrastructure for industry and academic institutions collaboration. As emphasized by UNESCO and World Bank, investing in resilient digital education platforms fosters both innovation and improved learning outcomes.

Synthesis of the Reviewed Literature and Studies

Key insights

Connection of different themes

Research gaps



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Chapter 3

METHODOLOGY

Research Design

This study will follow a descriptive quantitative research design with a prototype-based evaluation approach to gather stakeholder perceptions on the existing traditional OJT processes, the proposed OJTPortal, its alignment with the specific SDGs, and its feasibility for implementation. A survey questionnaire will be used as the main data gathering tool for obtaining objective, numerical data that support the assessment of the system's potential value and relevance to its intended users.

Software Development Methodology

The development for the OJTPortal utilizes the Agile Methodology. It ensures flexibility in enhancement, along with stakeholder engagement, across the entire process. The process will begin with issue identification and teamwork with stakeholders through issue definition, for example, deficiencies in the process of OJT, deficits in using the communication tool, or monitoring issues among students. These findings are utilized to develop a prioritized product backlog, outlining major features such as student registration, task tracking, communication tools, and reporting features.

The project is segmented into iterative sprints of 2–4 weeks, each aimed at developing functional elements that are added to the system incrementally. When ready, the system is deployed step by step for easy adoption, after which training is conducted for users and ongoing support to rectify any post-deployment issues. Comprehensive documentation in the form of user manuals and technical guides is



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provided for upkeep and future development. Through this iterative and collaborative process, the OJTPortal remains up to date with stakeholder needs and effectively fulfills its designed purpose.

Sources of Data

The primary sources of data in this study will be selected stakeholders involved in the OJT process. These include the following: (1) 3rd year OJT students in ITECH who have undergone or currently undergoing OJT; (2) OJT advisers responsible for supervising students; (3) OJT coordinators or administrative staff who manage OJT documentation and processes; and (4) partner HTE representatives who accept and evaluate student trainees. Their responses will serve as the main basis for analyzing the current challenges in OJT, evaluating the proposed system, and assessing its perceived alignment with SDGs and readiness for implementation.

Research Instrument

A structured survey questionnaire will be used as the primary instrument in this study to collect quantitative data from stakeholders after the orientation to the OJTPortal prototype. The questionnaire will be divided into five main sections: (1) demographic profile that contains the role of the respondent (student, adviser, coordinator, HTE representative); (2) challenges in the traditional OJT process; (3) perceived effectiveness of the OJTPortal prototype; (4) system acceptance based on the specific ISO/IEC 25010 characteristics; and (5) perceived contribution to SDGs and implementation feasibility.

All closed-ended items will be rated on a 5-point Likert scale. The Likert-type scale will measure the respondents' agreement to a statement wherein the values of one (1) to five (5) corresponds to "Strongly disagree," "Disagree," "Neutral," "Agree," and "Strongly agree" respectively. The survey questionnaire will be connected to the statement of the problem when determining the challenges, perceived effectiveness,



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system acceptance based on ISO 25010 standards, perceived contribution to SDGs, and implementation feasibility.

The overview about the study and the confidentiality of respondents' background information and survey answers will be mentioned before answering the questionnaire. The questions will be modified to fit the objectives of the study. The research instrument will be validated by a professor authorized to validate and make sure the questionnaire will be appropriate for the study.

In addition to the survey, the researchers will conduct preliminary interviews to gain contextual insights influencing the development of the questionnaire and the prototype itself. A semi-structured interview will be conducted with the OJT coordinator using prepared guide questions focused on common administrative challenges and workflow issues in the OJT process.

Furthermore, the researchers will hold informal interviews with selected OJT advisers and students to understand their personal experiences and pain points. These discussions will be unstructured and conversational in nature to allow respondents to speak freely about their concerns.

Data Gathering Procedure

The researchers will utilize a systematic method for gathering data on the construction of the proposed system. The data collection process is divided in two main phases: preliminary interviews and prototype-based survey administration.

In the first phase, the researchers will conduct a preliminary interview with the PUP ITECH OJT Coordinator to gather insights into the administrative challenges in the current OJT process. The interview will follow the prepared questions focused on the existing procedures, issues, and requirements concerning OJT. Further information includes the needs of the university, faculty, and students, together with recommendations, protocols, and restrictions on third-party involvement



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of HTEs. To complement this, informal interviews will be carried out with selected OJT advisers and students to understand individual experiences and recurring concerns. To further specialize in the features of the system and make it fully in line with user expectations, researchers will construct a detailed questionnaire on the subject.

Starting the second phase of data gathering, the researchers will conduct prototype demonstrations for the target respondents—students, advisers, coordinators, and HTE representatives. After the system walkthrough, respondents will be asked to complete a structured survey questionnaire evaluating the prototype's effectiveness, quality, and relevance to their needs. The researchers will distribute the survey online through Google Forms. Before participation, respondents will be informed on the on the study's purpose and asked to give informed consent.

The researchers will organize and analyze the data collected from interviews and questionnaires using appropriate descriptive and inferential statistical tools. The responses will give an overall picture of the requirements and feedback required to develop, design, and test the proposed system effectively.

Ethical Considerations

This research aims to enhance the management of On-the-Job Training programs through the development of the OJT Portal at the Polytechnic University of the Philippines. To ensure the protection of participants, informed consent will be obtained from all respondents. Participants will be fully informed about the study's purpose, data collection methods, and their right to withdraw at any time without penalty.

All personal data, including names and contact information, will be safely stored because confidentiality is of utmost importance. The data will be anonymized



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during analysis to protect participants' privacy. Data will also be collected using neutral, structured and impartial, allowing for accurate and reliable results.

The research will adhere to the ethical guidelines set forth by the Polytechnic University of the Philippines, which includes obtaining necessary approvals and undergoing an ethics review process. Following these standards guarantees the validity and integrity of the study while protecting the rights and privacy of all participants.

Data Case Analysis

The OJTPortal system manages essential and confidential data related to the OJT processes. This includes student records, OJT requirements, compiled portfolio, progress updates, evaluations, and feedback from involved stakeholders. The primary users of the systems are students, OJT advisers, OJT coordinators, and partner HTE representatives, each interacting with the system through specific modules.

The system collects input data such as ... These inputs will be processed and stored in internal databases are will be used to generate outputs such as progress reports, evaluation summaries, compliance checklists, ...

To visualize how data flows within the system, ...

User registration and login -> student information database

OJT document submission -> OJT documents repository

Progress monitoring

Feedback, evaluation and report generation -> evaluation records & feedback logs



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What data the system handles

Who provides the data (users/stakeholders)

What happens to the data (processes and outputs)

How it is represented in DFD

Statistical Treatment

Proposed System Architecture



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Chapter 4
RESULTS AND DISCUSSION



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Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS



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References